Regression Diagnostics on Cryptocurrency Volatility

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April 4, 2025

Abstract

This paper investigates the determinants of Bitcoin volatility using multiple linear regression, with a particular emphasis on regression diagnostics to assess model validity. Daily price and volume data from Yahoo Finance, are analyzed alongside supplementary predictors such as S&P 500 returns and market sentiment from X posts. The study employs transformations, outlier detection, and variable selection to refine the model, while diagnostic tools—residual plots, Variance Inflation Factor (VIF), and the Durbin-Watson test—evaluate assumptions like linearity, homoscedasticity, and independence. Preliminary findings suggest trading volume as a key driver of volatility, though time-series effects challenge standard regression assumptions. This analysis bridges financial data with statistical methodology, offering insights into cryptocurrency market dynamics.

1 Introduction

Cryptocurrency markets, exemplified by Bitcoin, exhibit extreme price volatility, making them a compelling subject for statistical analysis. Unlike traditional assets, cryptocurrencies lack centralized regulation, leading to rapid price swings influenced by trading activity, investor sentiment, and macroeconomic conditions. This project applies multiple linear regression to model Bitcoin volatility, defined as daily logarithmic returns, using predictors such as trading volume and market indices. However, financial data often violates regression assumptions (e.g., homoscedasticity, independence), necessitating a focus on diagnostics to ensure model reliability.

This study aligns with the objectives of *Statistics in Applications II* by exploring regression techniques—variable transformations, multicollinearity checks, and outlier handling—while emphasizing diagnostics as outlined in Weisberg (2014). The paper proceeds as follows: Section 2 details the data and methodology, Section 3 outlines the planned modeling process and anticipated results, and Section 4 concludes with implications and future directions.

2 Data and Approach

2.1 Data Collection

The primary dataset comprises daily Bitcoin (BTC-USD) price and trading volume data from Yahoo Finance, collected via Python's `yfinance` library. Volatility, the response variable, is calculated as $\log\left(\frac{\text{Price}_t}{\text{Price}_{t-1}}\right)$, where Price_t is the closing price on day t. Additional predictors include:

- *Trading Volume*: Daily USD volume from Yahoo Finance.

- *S&P 500 Returns*: Log returns of the S&P 500 index ($^{G}SPC), asaproxyformarketconditions.Market\ Se$
$Optionalsentimentscoresfrom X posts, derivedviakeywordanalysis(e.g., \backslash Bitcoin" mentions), iftimep$
Data sources are accessible at `https://finance.yahoo.com/quote/BTC-USD/history/` and `https://finance.yahoo.com/quote/^GSPC/history/`.

2.2 Data Preprocessing

Missing values will be addressed via listwise deletion, assuming randomness in gaps. Outliers, expected due to market shocks (e.g., flash crashes), will be identified using Cook's Distance and potentially winsorized. Predictors will be standardized if scale differences (e.g., volume vs. returns) affect model stability.

2.3 Model Specification

The initial model is a multiple linear regression:

$$Y_{(n\times1)} = X_{(n\times p)}\beta_{(p\times1)} + \epsilon_{(n\times1)}, \quad \epsilon \sim N(0, \sigma^2 I),$$

where Y is volatility, X includes trading volume, S&P 500 returns, and sentiment (if included), and β represents coefficients. The analysis will proceed iteratively, refining the model through diagnostics and adjustments.

2.4 Modeling Process

The fitting process follows these steps:

1. *Step 0: Initial Fit* — Regress volatility on all predictors. Assess baseline R^2 and coefficient significance.
2. *Step 1: Diagnostics* — Examine residual plots for nonlinearity or heteroscedasticity, Q-Q plots for normality, and Durbin-Watson for autocorrelation. Compute VIF to detect multicollinearity (e.g., volume vs. sentiment).
3. *Step 2: Transformations* — If skewness or heteroscedasticity is evident, apply logarithmic transformations to predictors (e.g., volume) or use robust standard errors.
4. *Step 3: Outlier Handling* — Remove or adjust influential points identified via Cook's Distance, re-evaluating fit.
5. *Step 4: Variable Selection* — Apply stepwise regression (AIC/BIC) to reduce predictors, balancing parsimony and explanatory power.

Software tools include Python (`statsmodels`, `pandas`) for regression and diagnostics, with visualizations via `matplotlib`.

2.5 Potential Challenges

Financial time-series data may exhibit autocorrelation, detectable via Durbin-Watson values far from 2, suggesting lagged variables or time-series models as alternatives. Heteroscedasticity, common in volatile markets, may require robust methods. Collinearity between volume and sentiment could inflate VIF, necessitating variable reduction. These issues will be documented and addressed, enhancing the diagnostic focus.

3 Preliminary Results and Interpretation

Exploratory analysis of Bitcoin data suggests trading volume strongly correlates with volatility, while S&P 500 returns may capture broader market influence. The initial model is expected to explain 60–70% of variance (R^2), though diagnostics may reveal violations like autocorrelation due to time dependencies. For example, a 1% increase in volume might increase volatility by 0.02 units, holding other factors constant, though this interpretation hinges on assumption validity. Final results will quantify predictor effects and diagnostic outcomes, potentially highlighting limitations of linear regression for financial data.

4 Conclusion

This project applies multiple linear regression to Bitcoin volatility, emphasizing diagnostics to validate assumptions in a challenging dataset. It bridges statistical theory with cryptocurrency analysis, offering insights into market drivers and model robustness. Future work could explore nonlinear or time-series models (e.g., GARCH) to address residual issues, enhancing predictive accuracy.

References

[1] Weisberg, S. (2014). *Applied Linear Regression*. Wiley, 4th edition.